Computational analysis of antimicrobial peptides identified in the coral *Acropora digitifera* as a possible alternative for antimicrobial therapeutics.

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Introduction.

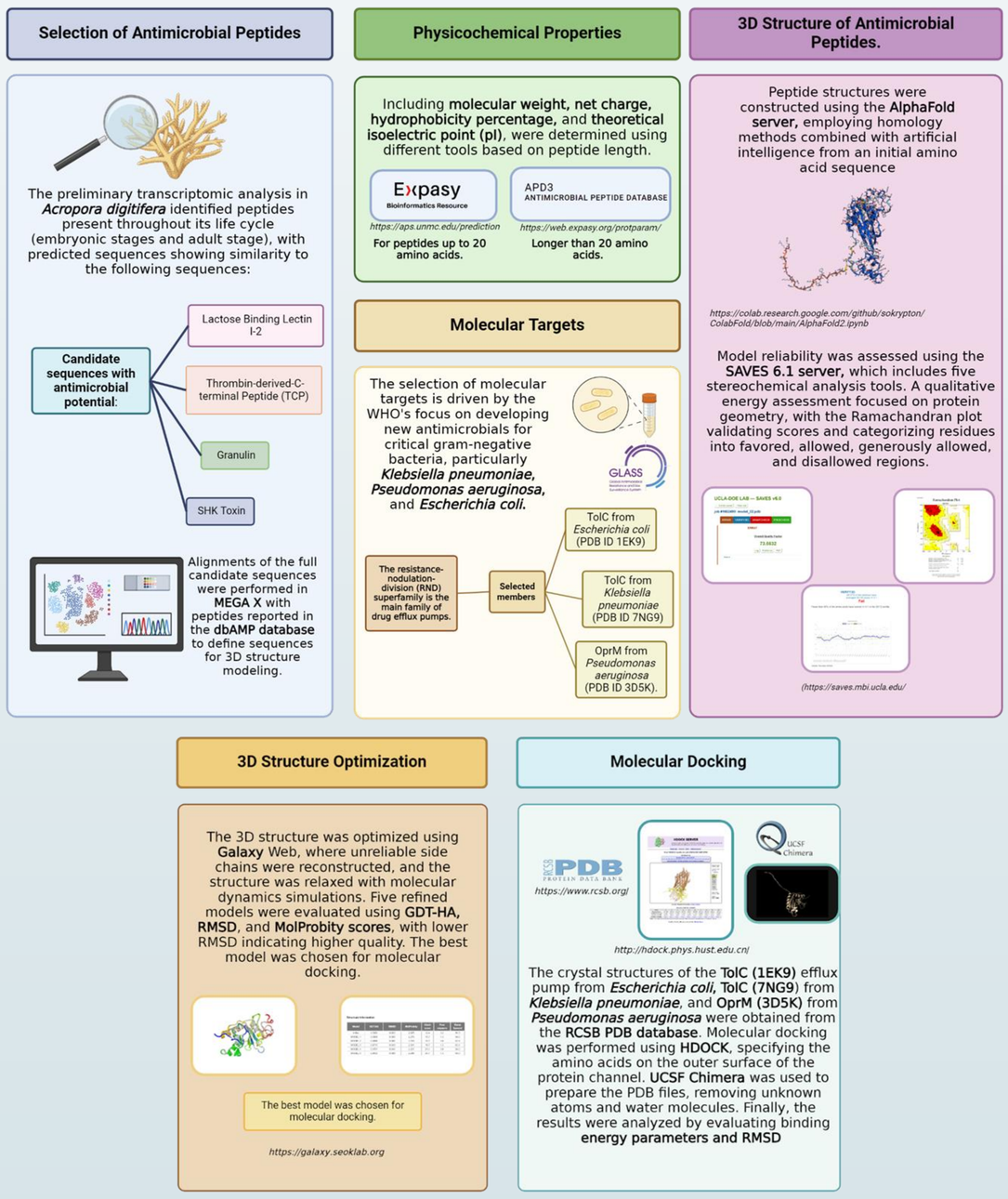
Antimicrobial resistance (AMR) is a serious threat to global public health, exacerbated by the overuse of antibiotics and the lack of new therapeutic alternatives. The GLASS 2022 report indicates a significant increase in resistant microorganisms, with projections of up to 10 million annual deaths from resistant infections by 2050 (*Global Antimicrobial Resistance and Use Surveillance System (GLASS) Report 2022*, 2022). The pharmaceutical industry has shown little interest in developing new antibiotics due to their low profitability and long development times, perpetuating the problem (Cook & Wright, 2022; Vanegas Múnera & Jiménez Quiceno, 2020).

Efflux pumps are a key factor in antimicrobial resistance (AMR), allowing bacteria to eliminate antibiotics and toxic substances, which contributes to multidrug resistance (Alenazy, 2022). These mechanisms are especially common in Gram-negative pathogens, where efflux pumps, such as those in the RND superfamily, play a crucial role in resistance to multiple drugs (Du et al., 2018). These pumps are homotrimeric proteins in the outer membrane of bacteria and associate with six membrane fusion proteins (MFP), forming a tripartite complex through relatively weak tip-to-tip interactions that interact with outer membrane proteins (OMP). It has been shown that the efflux pumps TolC of *E. coli*, OprM of *P. aeruginosa*, and TolC of *K. pneumoniae* have this structure (Zwama & Nishino, 2021).

Considering AMR mechanisms, antimicrobial peptides (AMPs) emerge as a promising alternative. Identified in prokaryotes and eukaryotes, AMPs have a broad spectrum of activity with a low potential for inducing resistance (Bahar & Ren, 2013; Olascoaga-Del Angel et al., 2018). In particular, cnidarians, marine invertebrates with an innate immune system, produce AMPs that could be effective against multidrug-resistant infections. Moreover, these organisms have genetic and molecular structures that are promising for investigating new antimicrobial agents, due to their defense mechanisms similar to the human immune system (Leal et al., 2022).

The development of in silico analysis strategies, such as molecular docking, allows the exploration of the therapeutic potential of AMPs by identifying possible interactions between peptides and resistant pathogens (Pagadala et al., 2017). In this regard, peptides identified in the transcriptome of the coral *Acropora digitifera* have been evaluated by computational methods as a potential alternative for antimicrobial therapy.

Methodology.



Results.

Physicochemical properties: Physicochemical properties, including molecular weight, net charge, hydrophobicity percentage, and theoretical isoelectric point (pI), were determined using the APD 3 database and Expasy Tools (Table 1).

Peptide.	Aminoacids.	Molecular weight.	Net charge.	Hydrophobicity percentages.	Theoretical pI.
LBL I-2.	42.	4837,63	4	43%	10,02
Granulin.	43.	4320,9	-4,75	40%	3,71
TCP.	40.	4265,96	0	45%	6,11
SHK.	55.	6176,91	2,5	29%	8,81

Table 1. Physicochemical Properties.

Three-dimensional structure: Four candidate sequences of antimicrobial peptides identified in the transcriptomes of the coral *Acropora digitifera* were modeled with Alpha Fold (reference). The sequence Lactose Binding Lecitin I-2 consists of two α-helices joined by a single chain of amino acids. The Trombin Derived C-terminal peptide sequence presents a mixed structure, composed of two β-sheets and one α-helix. The Granulin sequence consists of a structure of four β-sheets. Finally, the SHK toxin shows a mixed structure, formed by two β-sheets and one α-helix. (Figure 1).

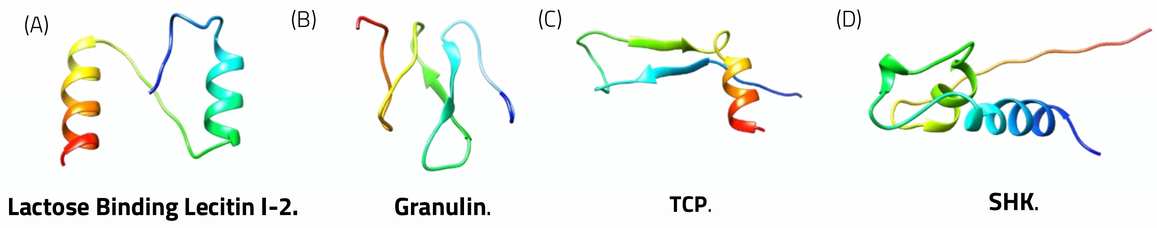


Figure 1. (A) α-Helix (B) β-Sheet (C) Combined structure of α-Helix and β-Sheet (D) Combined structure of α-Helix and β-Sheet.

Validation of the Three-Dimensional Structure: Using the SignalP 6.0 server, a qualitative analysis of the model's energy was conducted. The results were demonstrated through Ramachandran plots, which showed that between 95% and 100% of the residues were located in the favored and allowed regions (Table 2).

Peptide.	Favored Regions.	Allowed regions.	Generously allowed regions.	Disallowed regions.
LBL I-2.	100%	0%	0%	0%
Granulin.	93,3%	6,7%	0%	0%
TCP.	96,7%	3,3%	0%	0%
SHK.	97,6%	2,4%	0%	0%

Table 2. Percentages from the Ramachandran Plot analysis.

Molecular Docking: The molecular docking process was carried out between the candidate antimicrobial peptides and the RND-type efflux pumps of *E. Coli*, *P. Aeruginosa*, and *K. Pneumoniae* (Figure 2). The binding energy values and the root mean square deviation (RMSD) demonstrate that the peptides bind to the efflux pumps in the portion that integrates into the outer membrane (Table 3). Both in Figure 2 and Table 3, the best results for each peptide are observed. The efflux pump that presented the greatest affinity with the peptides was that of *E. coli*, which had the best results in binding energy and low to intermediate RMSD for three of the four peptides, indicating stable bonds.

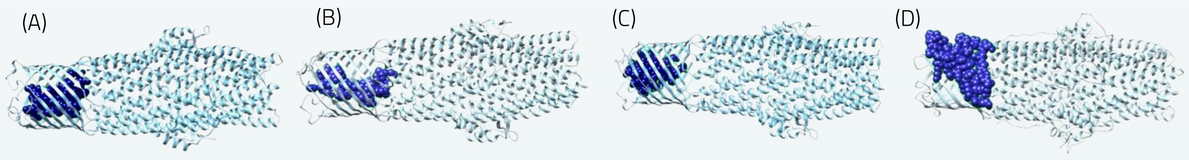


Figure 2. (A) LBL I-2 vs TolC *E. Coli* (B) TCP vs TolC *E. Coli* (C) Granulin vs TolC *E. Coli* (D) SHK vs OprM *P. Aeruginosa.*

Molecular Docking.	Binding Energy (Kcal/mol).	Ligand RMSD (Å).	ΔG (Kcal/mol).
LBL I-2 vs TolC *E. Coli*	-205,58	61,04	-10,1
TCP vs TolC *E. Coli*	-270,2	58,61	-13,2
Granulin vs TolC *E. Coli*	-189,60	57,38	-9,5
SHK vs OprM *P. Aeruginosa*	-285,72	16, 33	-10,1

Table 3. Binding Energies, RMSD and ΔG of the Molecular Dockings.

Discussion/Conclusion

AMR is one of the main challenges for global public health, intensified by the indiscriminate use of antibiotics and the lack of new alternatives. The growing threat of resistant infections highlights the need to develop effective alternatives, especially against Gram-negative bacteria such as *K. pneumoniae, P. aeruginosa,* and *E. coli* (Global Antimicrobial Resistance and Use Surveillance System (GLASS) Report 2022, 2022). In this context, AMPs derived from marine organisms, especially from the phylum Cnidaria, stand out as promising therapeutic options (Leal, Múnera, et al., 2022).

The selected AMPs exhibit notable structural diversity, with α-helices, β-sheets, or combinations of both (Koehbach & Craik, 2019). To evaluate the quality of ligand-protein interactions, binding energy and RMSD are used. RMSD measures the similarity between the actual and calculated positions of a ligand, with lower scores indicating a more stable binding. Regarding binding energy, more negative values reflect greater affinity in the docking (López-Camacho et al., 2016). Molecular docking results indicate stable interactions between four AMPs from *Acropora digitifera* and the TolC and OprM channels of RND-type efflux pumps, with the SHK peptide showing the best binding energy (-285.72) and low RMSD (16.33) when interacting with the OprM channel of *P. Auruginosa,* suggesting a high potential to inhibit the function of efflux pumps. Molecular docking between the OprM channel of *P. aeruginosa* and antimicrobial peptides (AMPs) from the human microbiome suggests that these AMPs can inhibit OprM activity (Mulpuru & Mishra, 2022). Additionally, the interaction between AMPs from the APD3 database and the TolC channel of Candidatus Liberibacter asiaticus reduced the surface area at the efflux pump exit, which could have affected its function, a phenomenon also observed with the AMPs analyzed in this study (Wang et al., 2022).

In conclusion, the interaction between antimicrobial peptides (AMPs) and RND-type efflux pumps is crucial for developing new therapies against antimicrobial resistance. In particular, marine AMPs stand out for their broad diversity and various mechanisms of action, making them a promising option. These preliminary studies not only enhance our understanding of the interaction between AMPs and their target proteins but also lay the foundation for exploring effective alternatives against resistant pathogens.

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